

碩專班電子學實驗 課前教學

課程教師：林尚亭 副教授

課程助教：林承恩

a

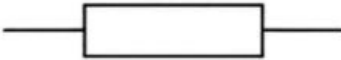
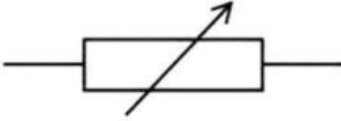
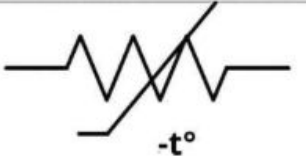
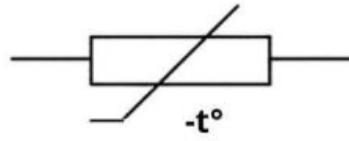
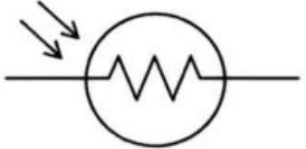
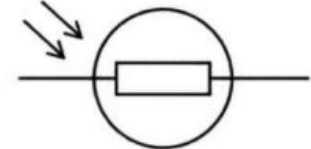
Tinkercad 簡介

常用元件

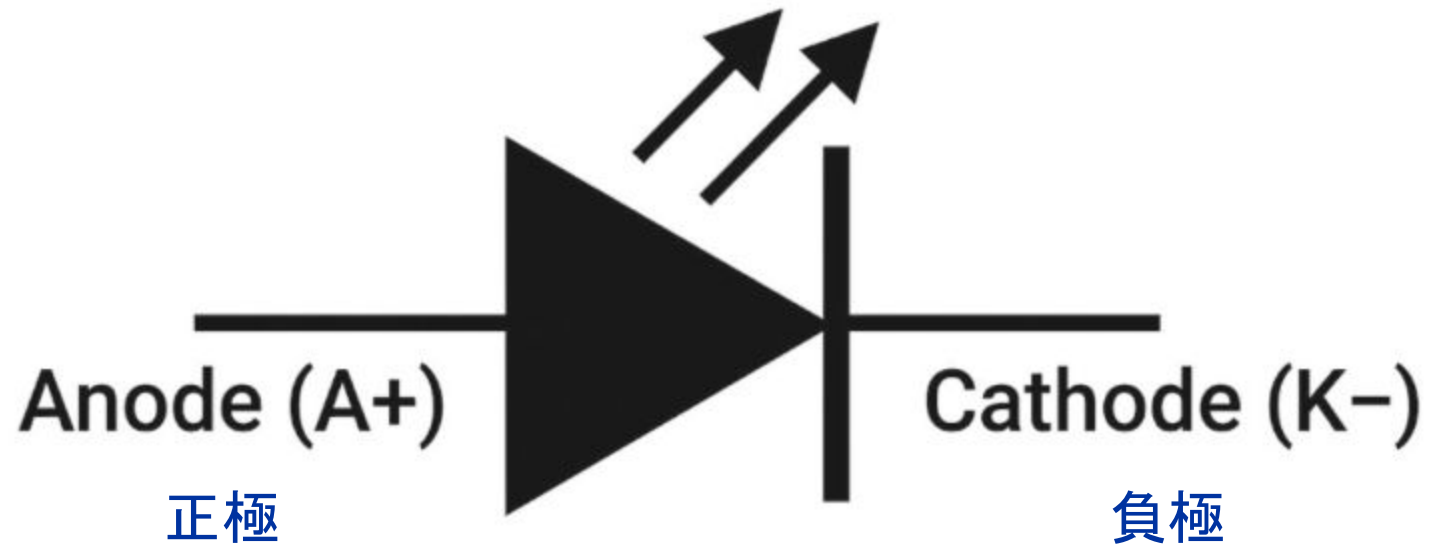


電路符號 - 電阻

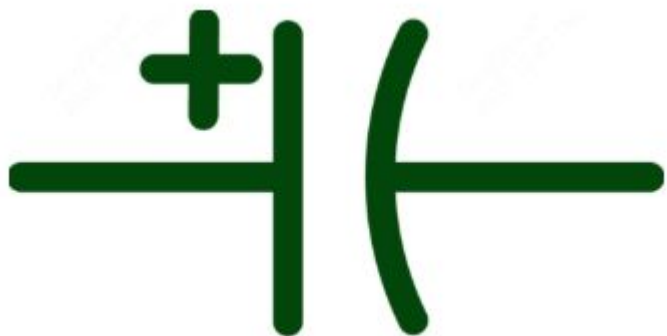
RESISTOR SYMBOLS – ANSI VS IEC

RESISTOR TYPE	ANSI / IEEE (ZIG-ZAG STYLE)	IEC (RECTANGLE STYLE)
1 Fixed Resistor (Constant Resistance)		
2 Variable Resistor (Potentiometer / Rheostat)		
3 Thermistor (Temperature Dependent)		
4 LDR (Light Dependent Resistor)		

電路符號 - LED



電路符號 - 電容



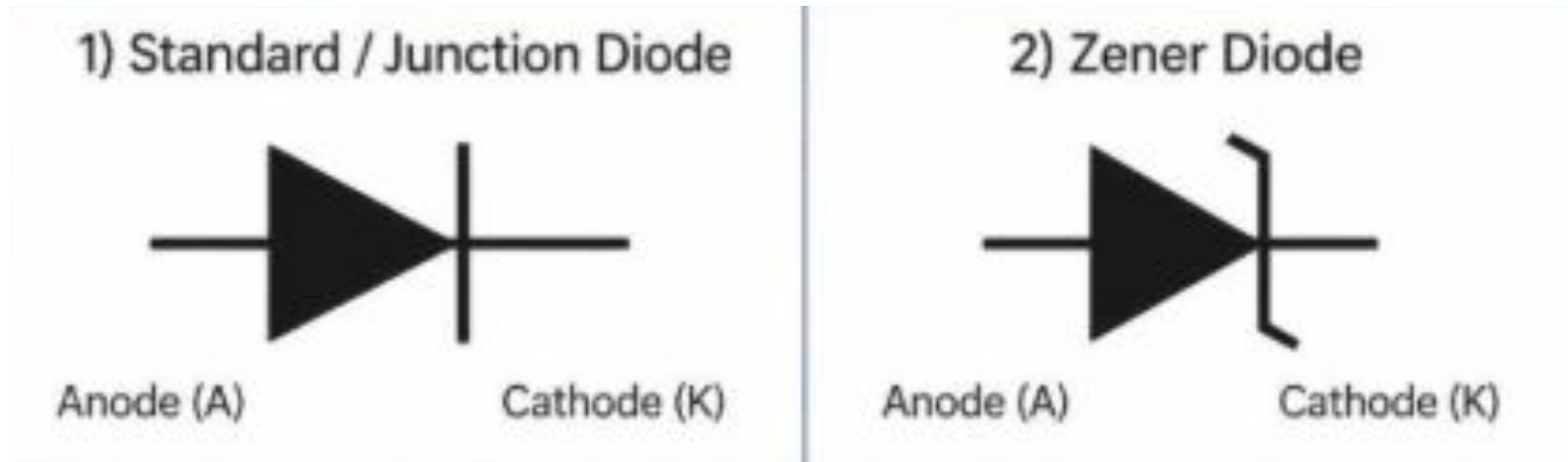
ANSI 風格極性電容符號

IEC 風格：外框方塊，於陽極處標示加號 (+)。



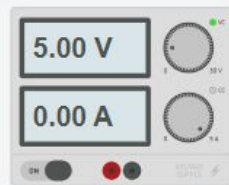
IEC 風格極性電容符號含加號

電路符號 - 二極體



齊納二極體

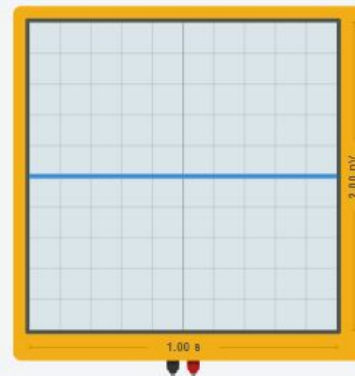
常用設備



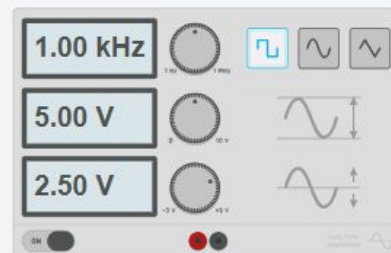
電源供應器



萬用電表
量測：電流、電壓、電阻

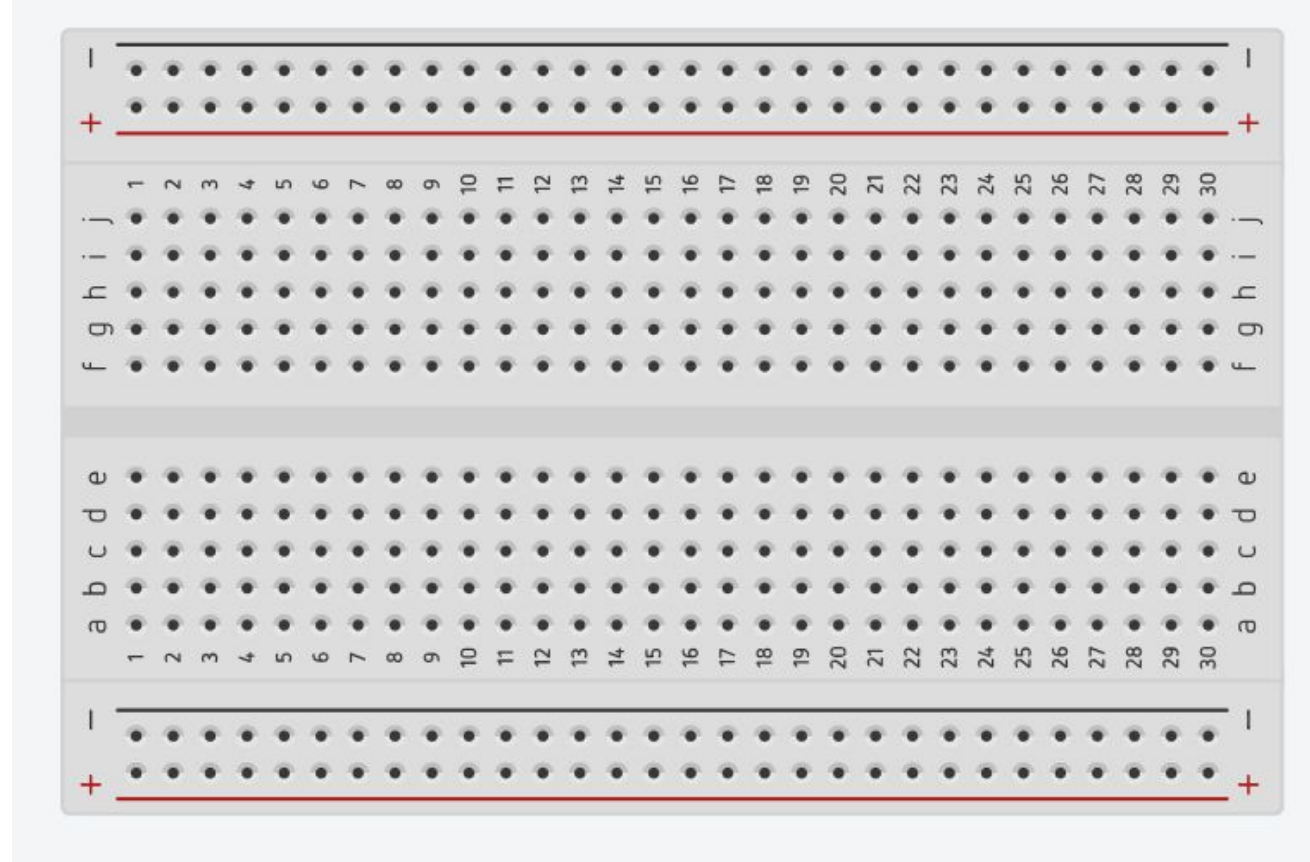


視波器



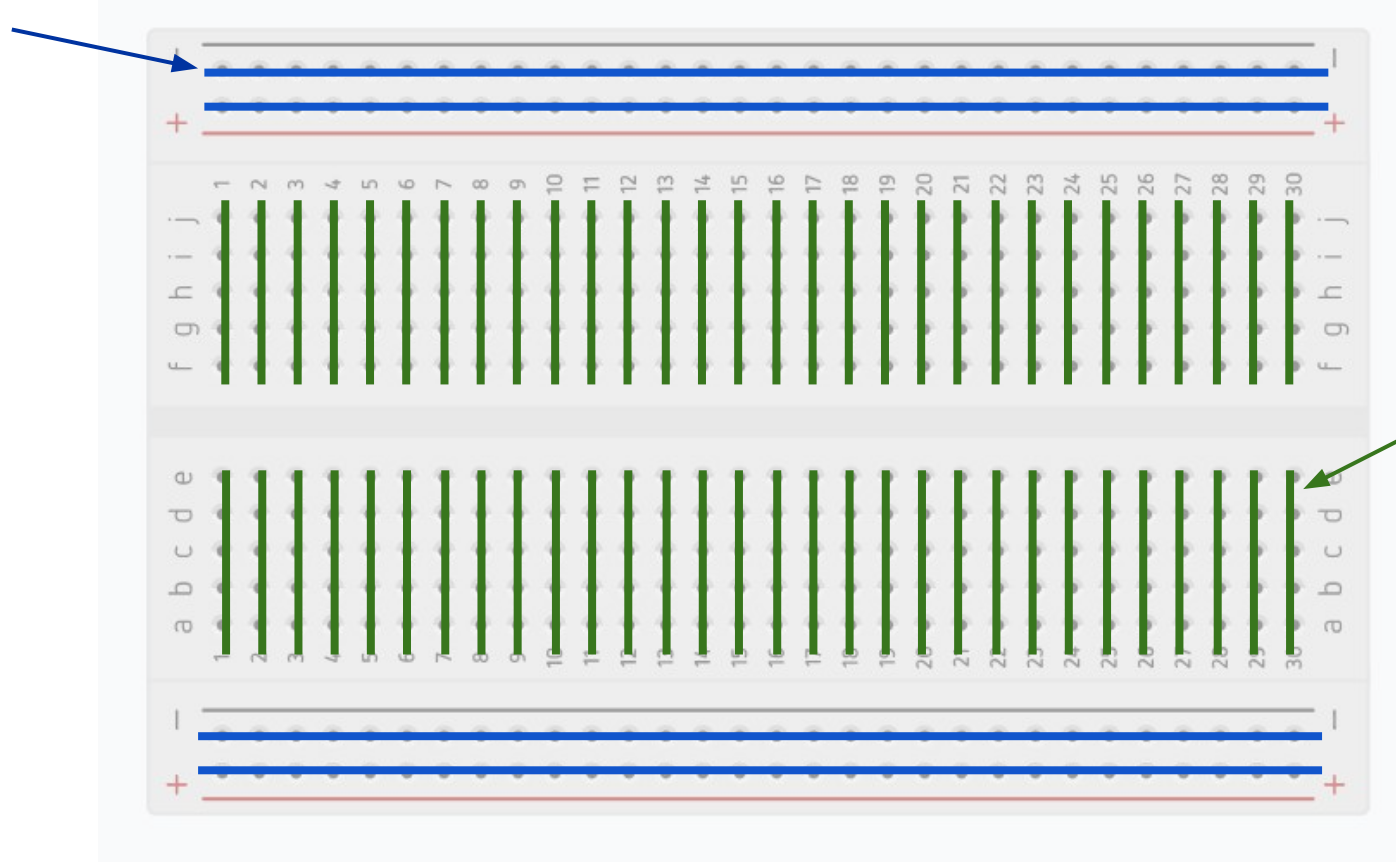
函數/訊號產生器

麵包版



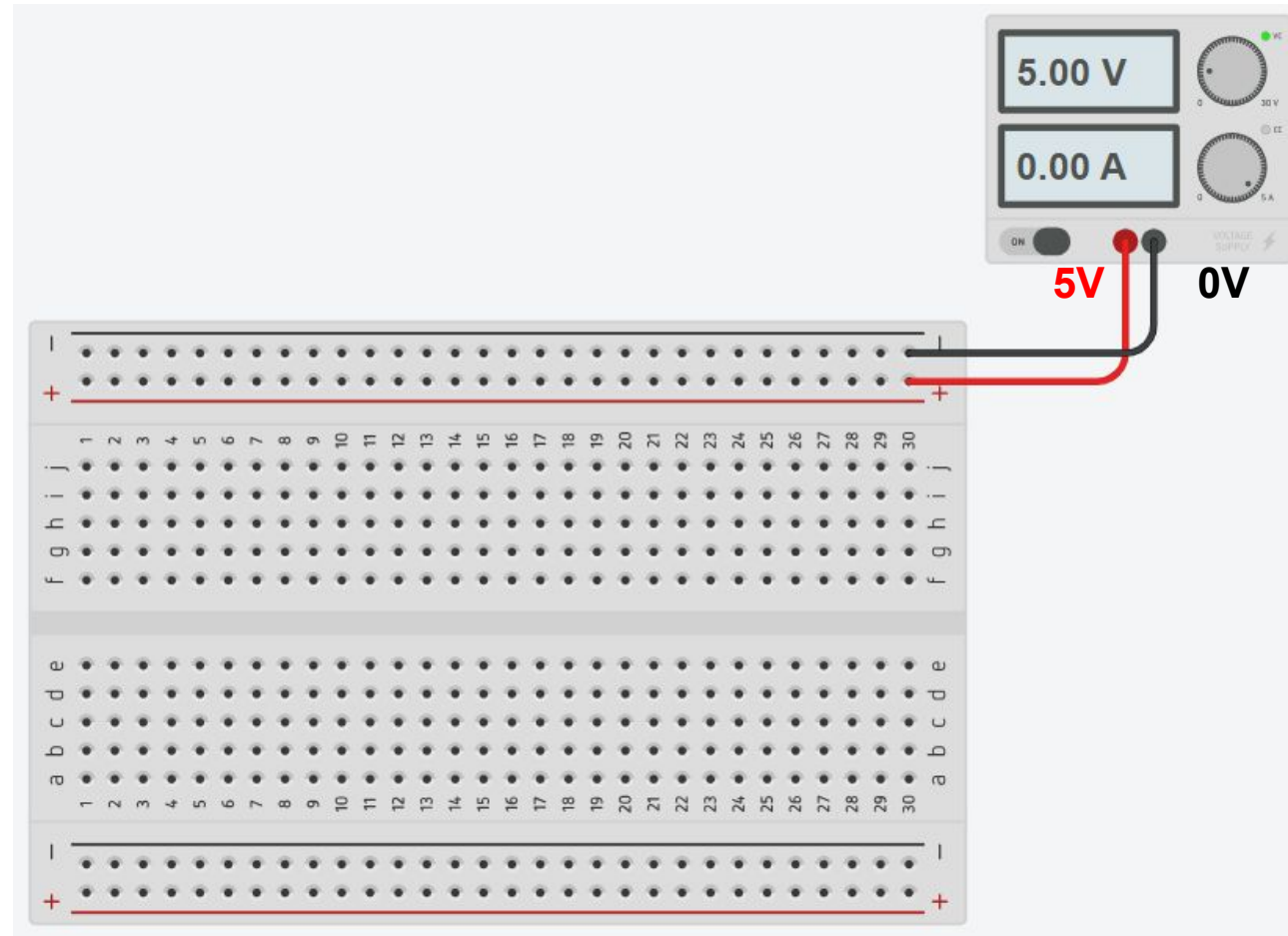
麵包版

在同一條導線上

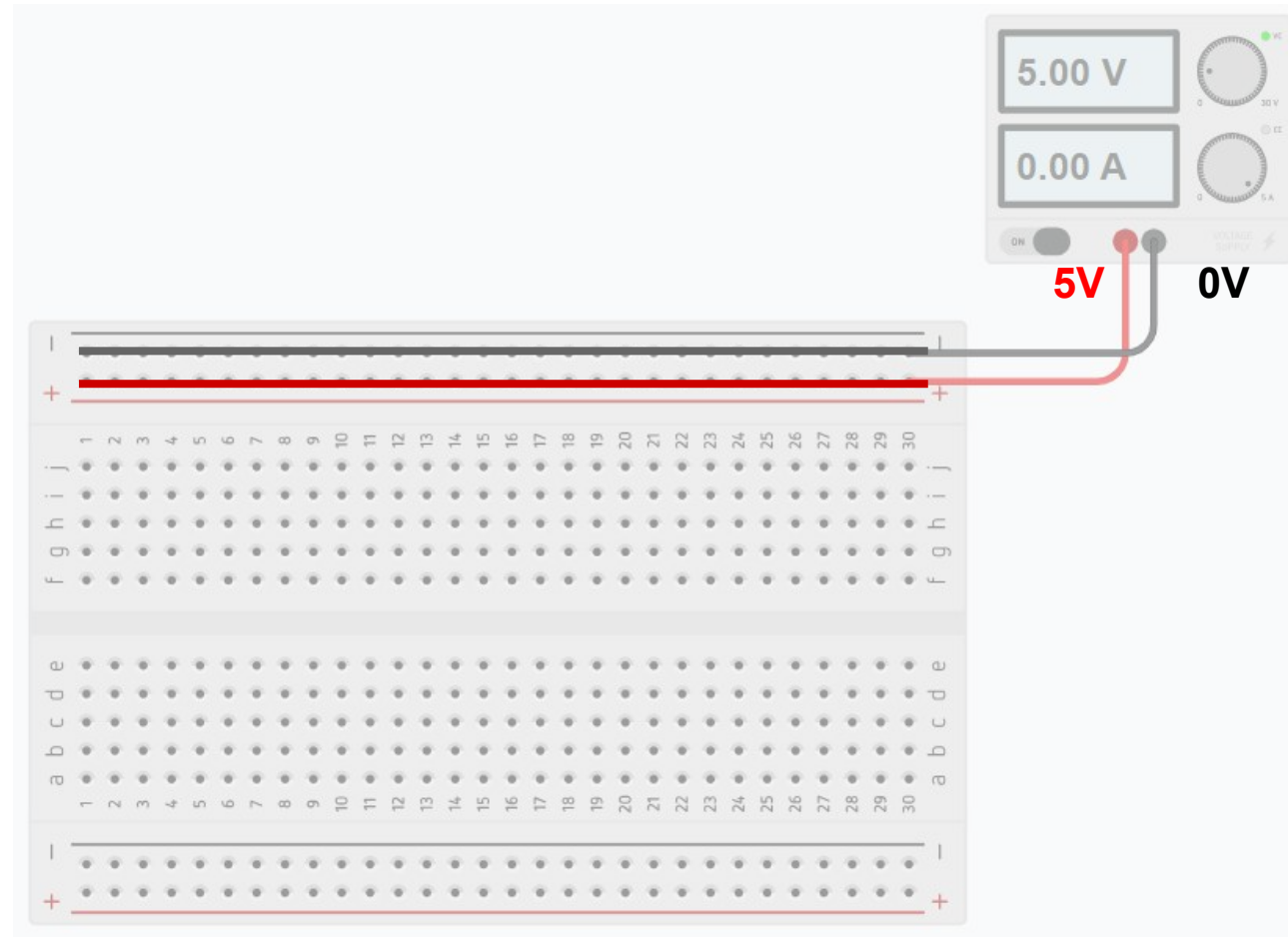


在同一條導線上

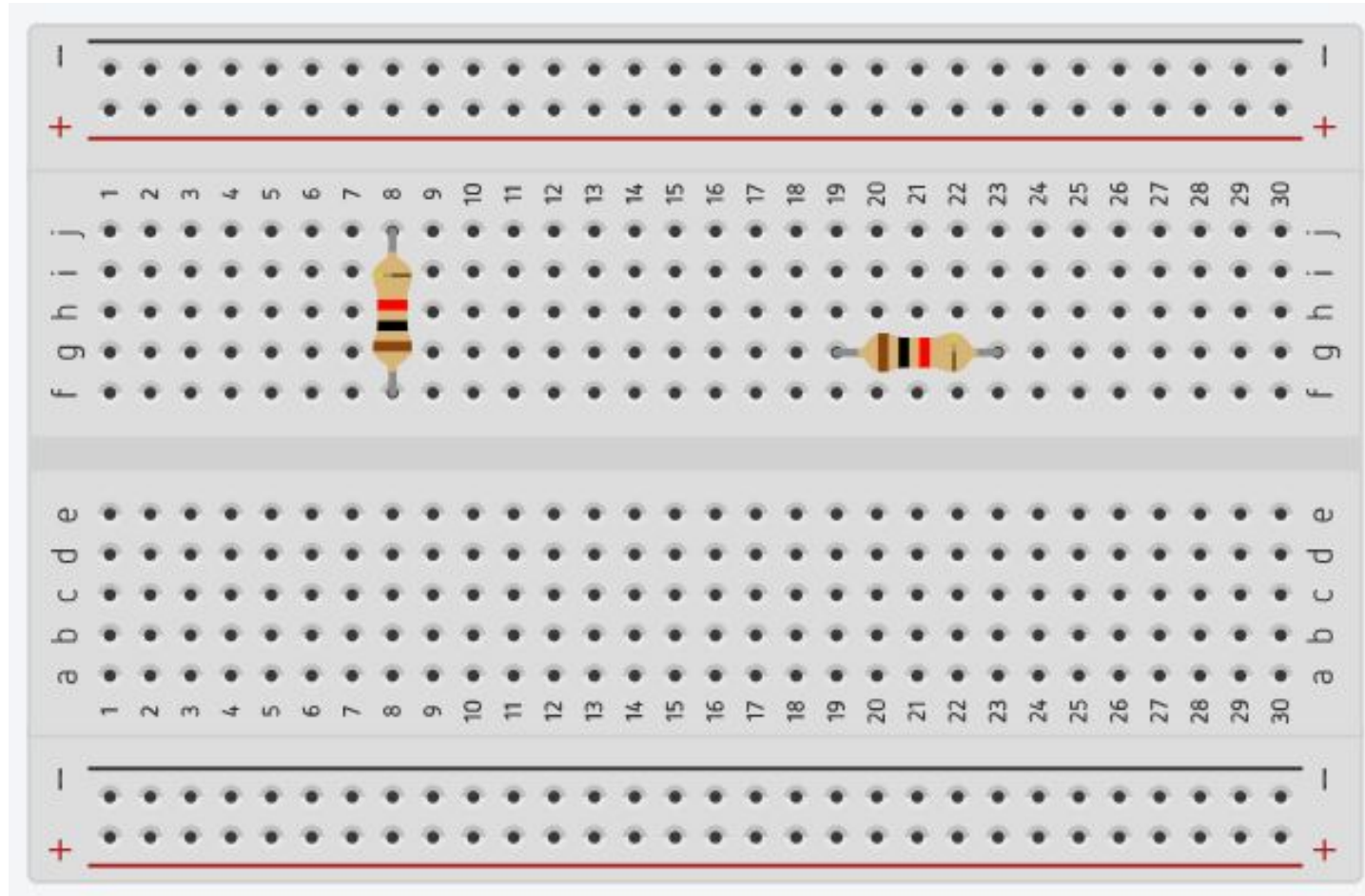
麵包版



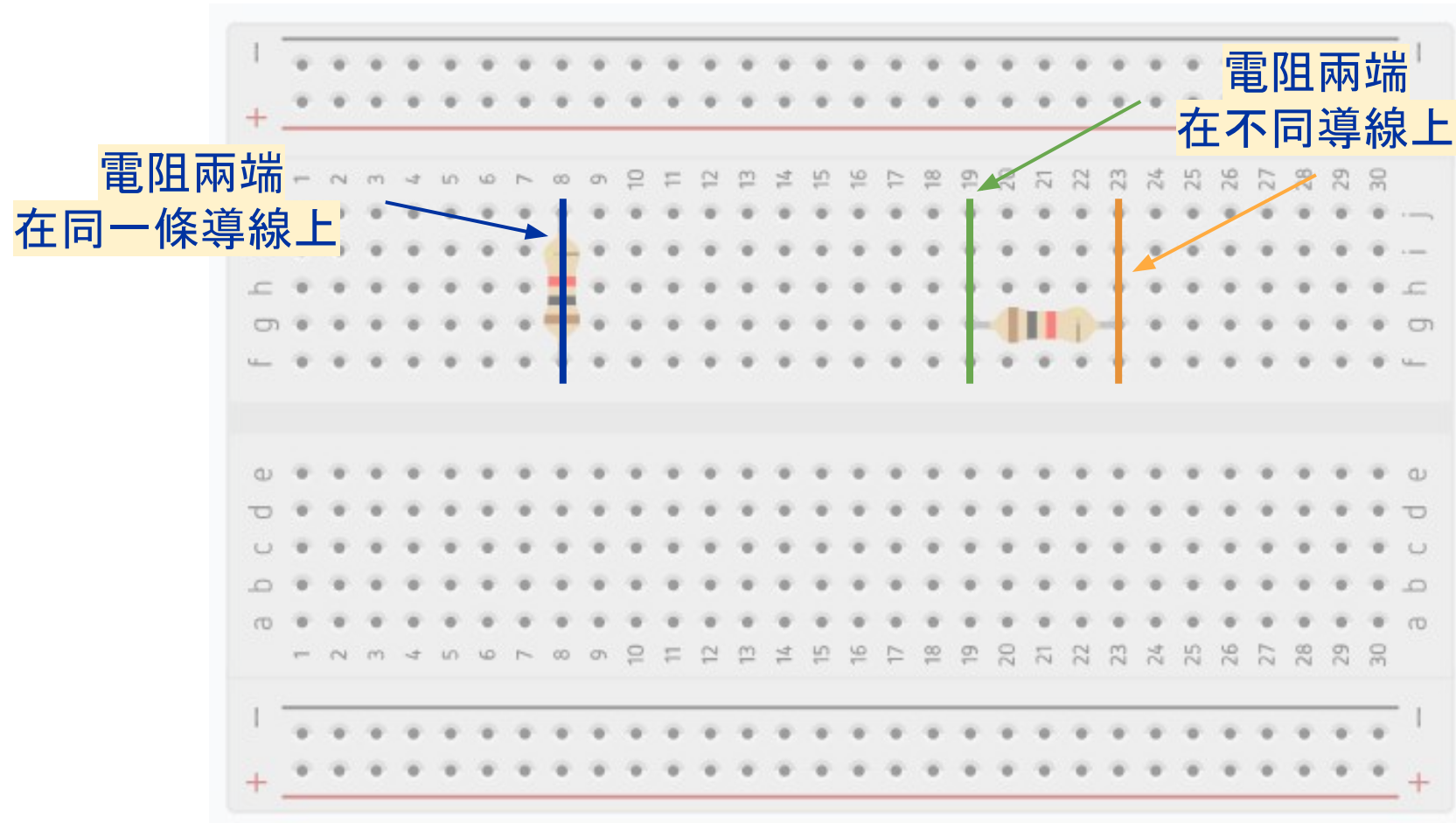
麵包版



麵包版



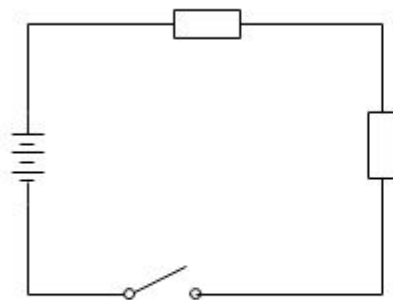
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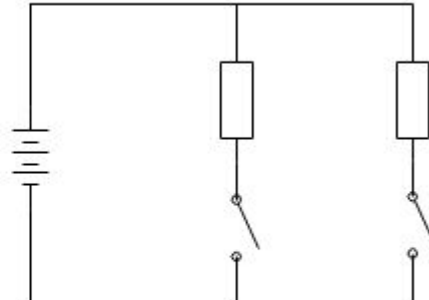
b

串聯 vs. 並聯

串聯 vs. 並聯



Series



Parallel

動腦時間

1. 使用萬用電表量測某電阻的**電流**時，應使電表與電阻【串聯 / 並聯】
2. 使用萬用電表量測某電阻的**電壓**時，應使電表與電阻【串聯 / 並聯】
3. 使用萬用電表量測某電阻的**電阻**時，應使電表與電阻【串聯 / 並聯】

動手時間

1. 使用 Tinkercad 完成下面的電路，並思考在實際電路板上應該怎麼接：

